

Developing a Martini gold model for use in biological contexts

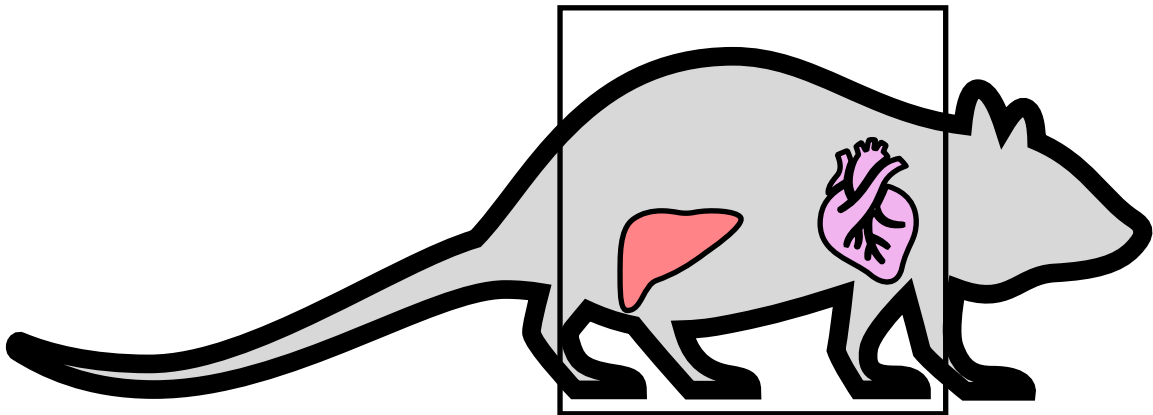
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Jahmal Ennis

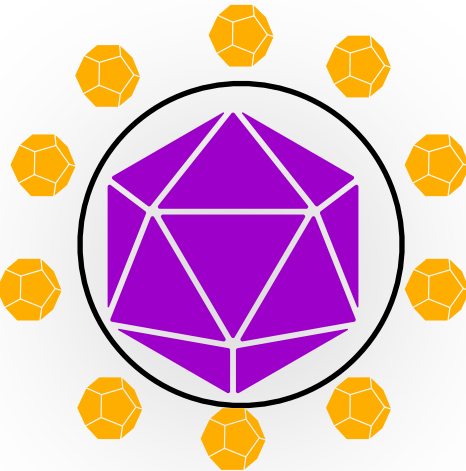
Background

What are the biomedical applications of gold?

Imaging



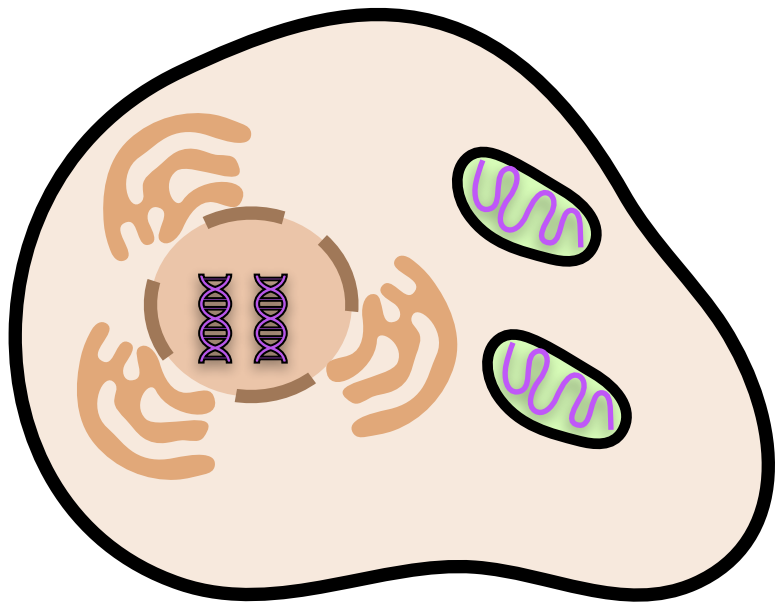
Diagnostics



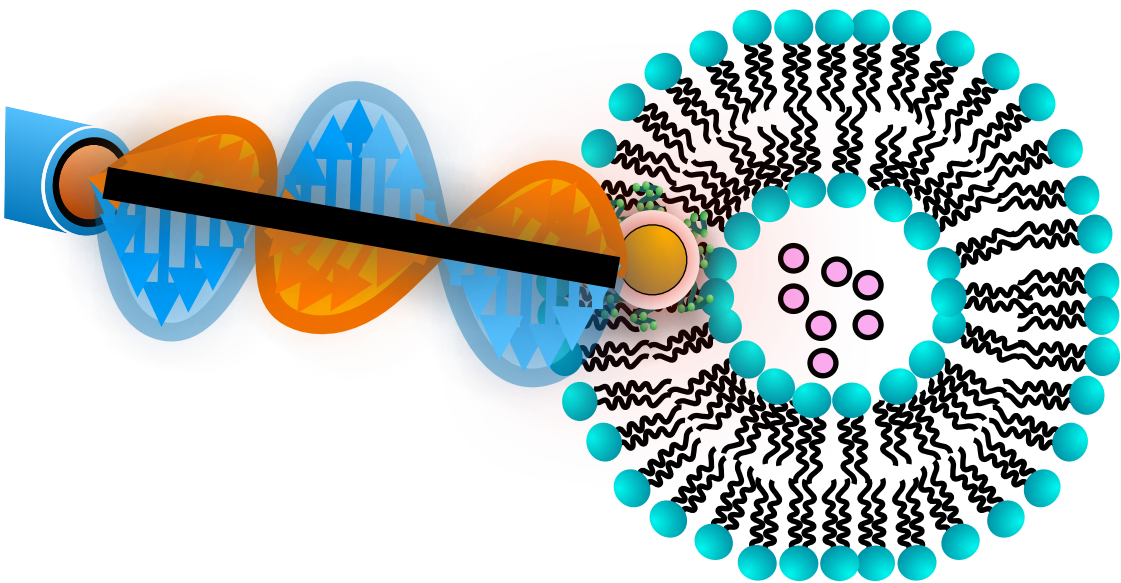
Gold Core

Movie was here

Therapy

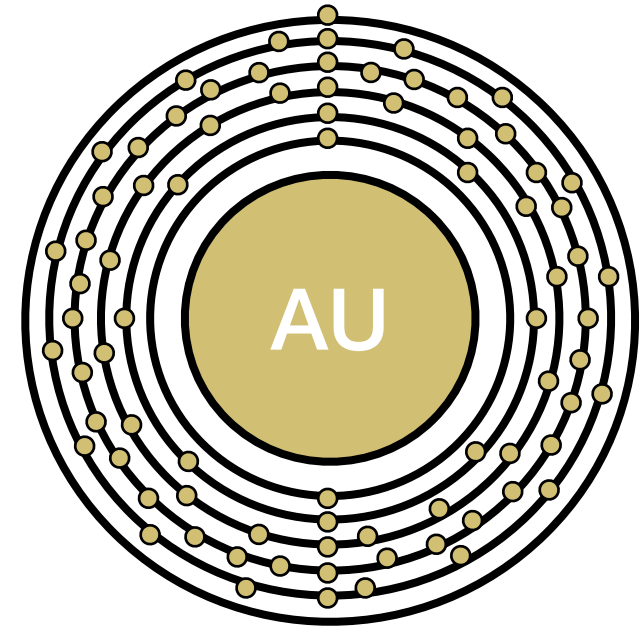


Ligands

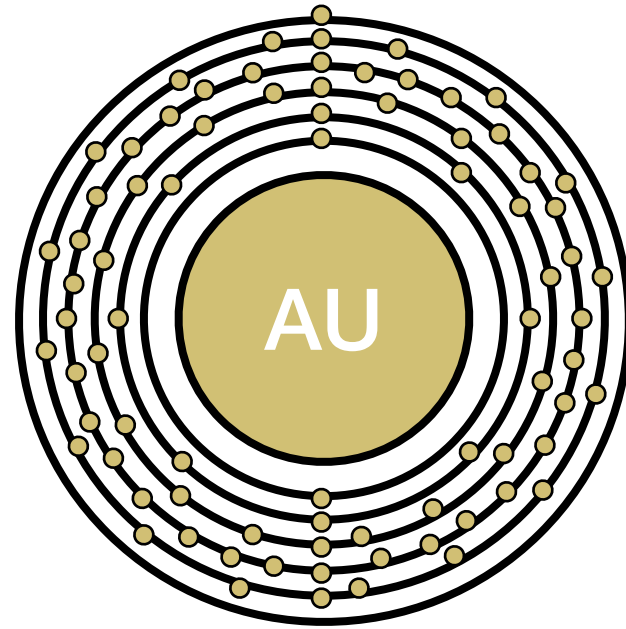
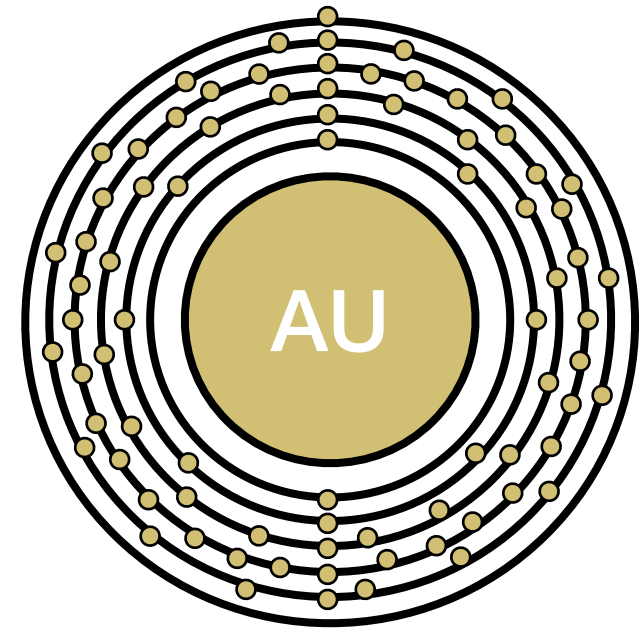


Drug Delivery

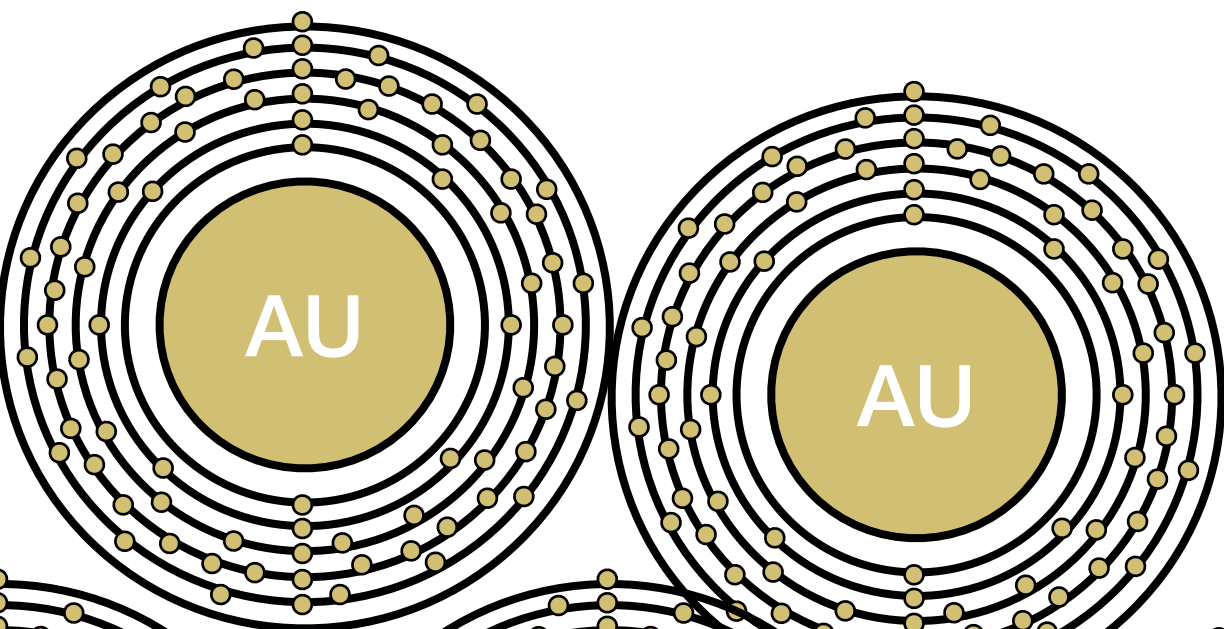
Properties of Nanoscale Gold



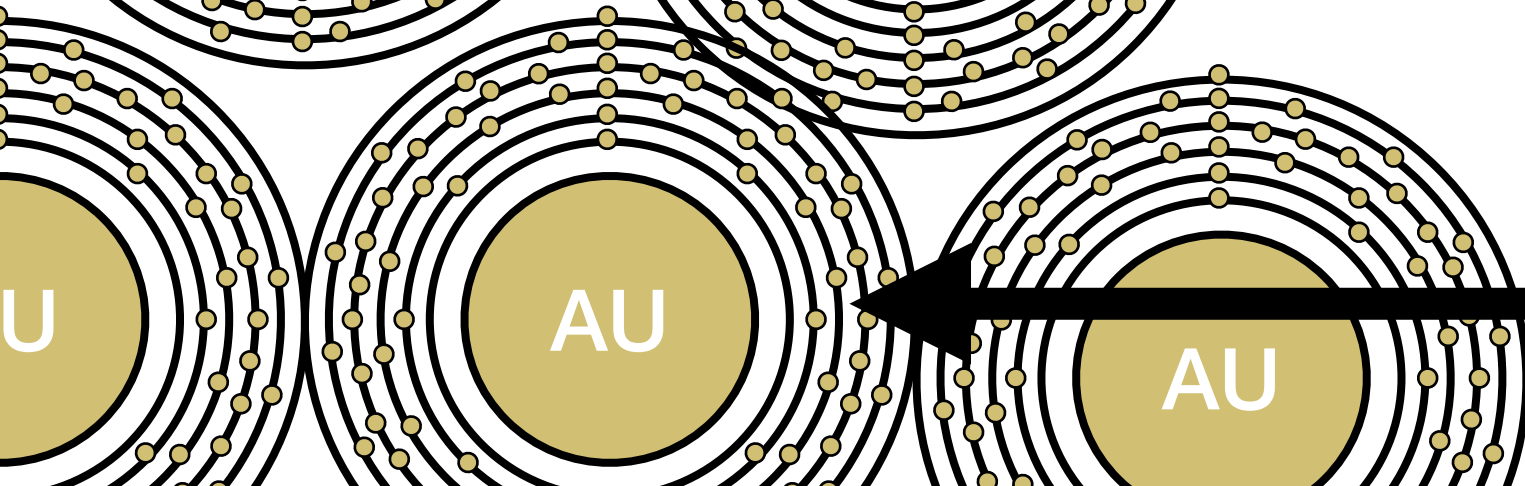
Large atoms with weak hold on their electrons



Form crystal lattices due to metal bonding



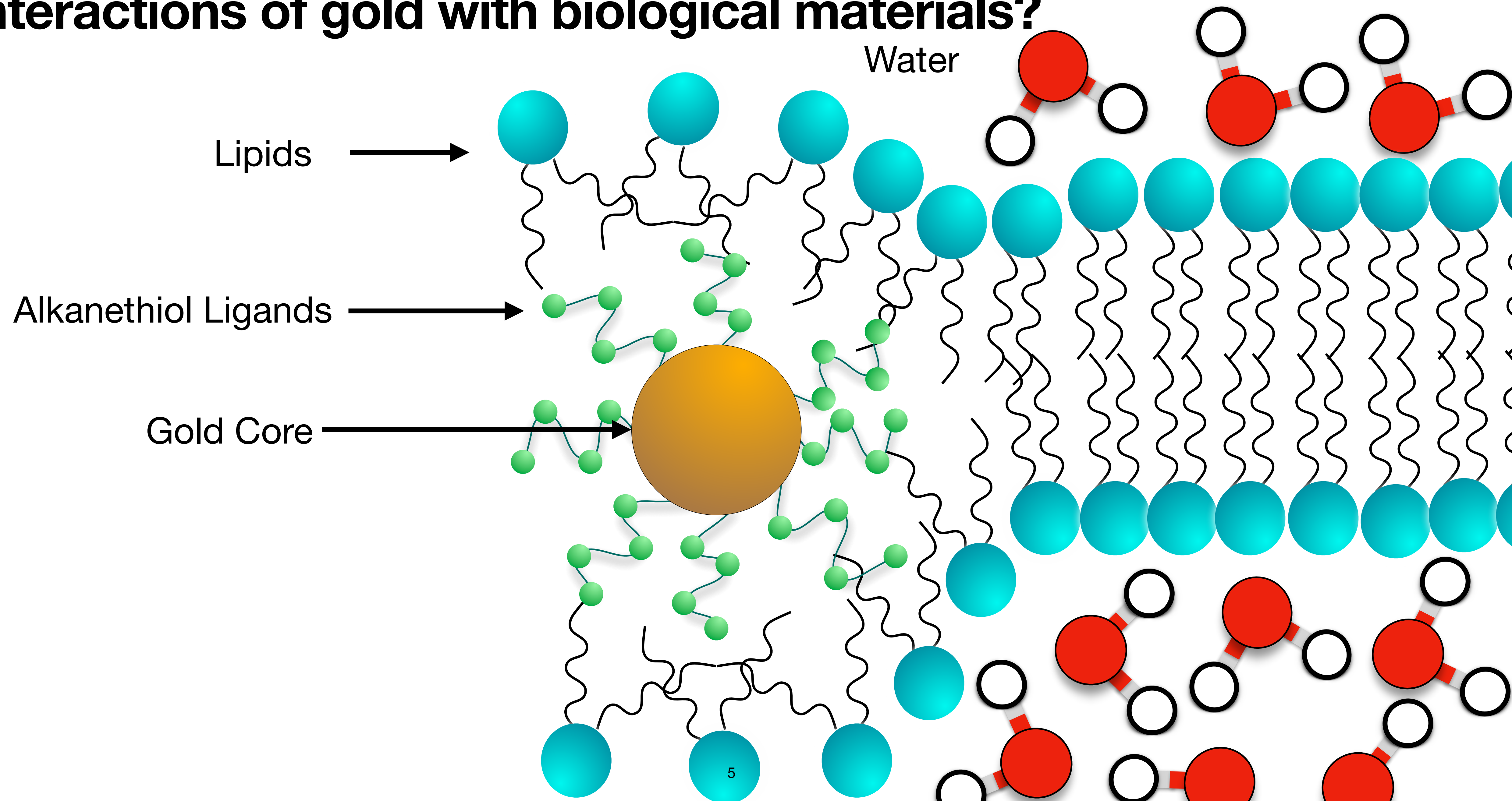
Frustrated



Satisfied

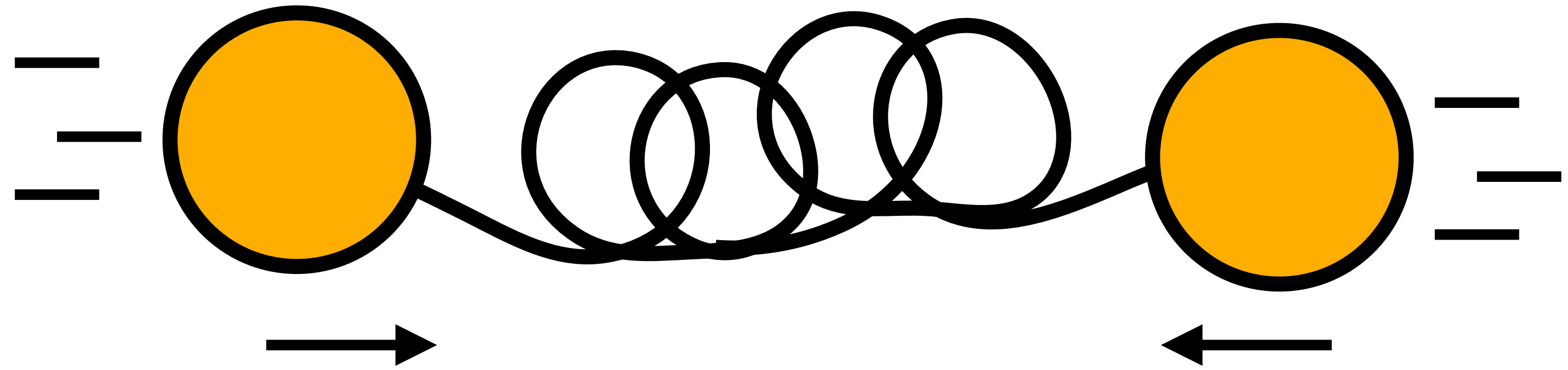
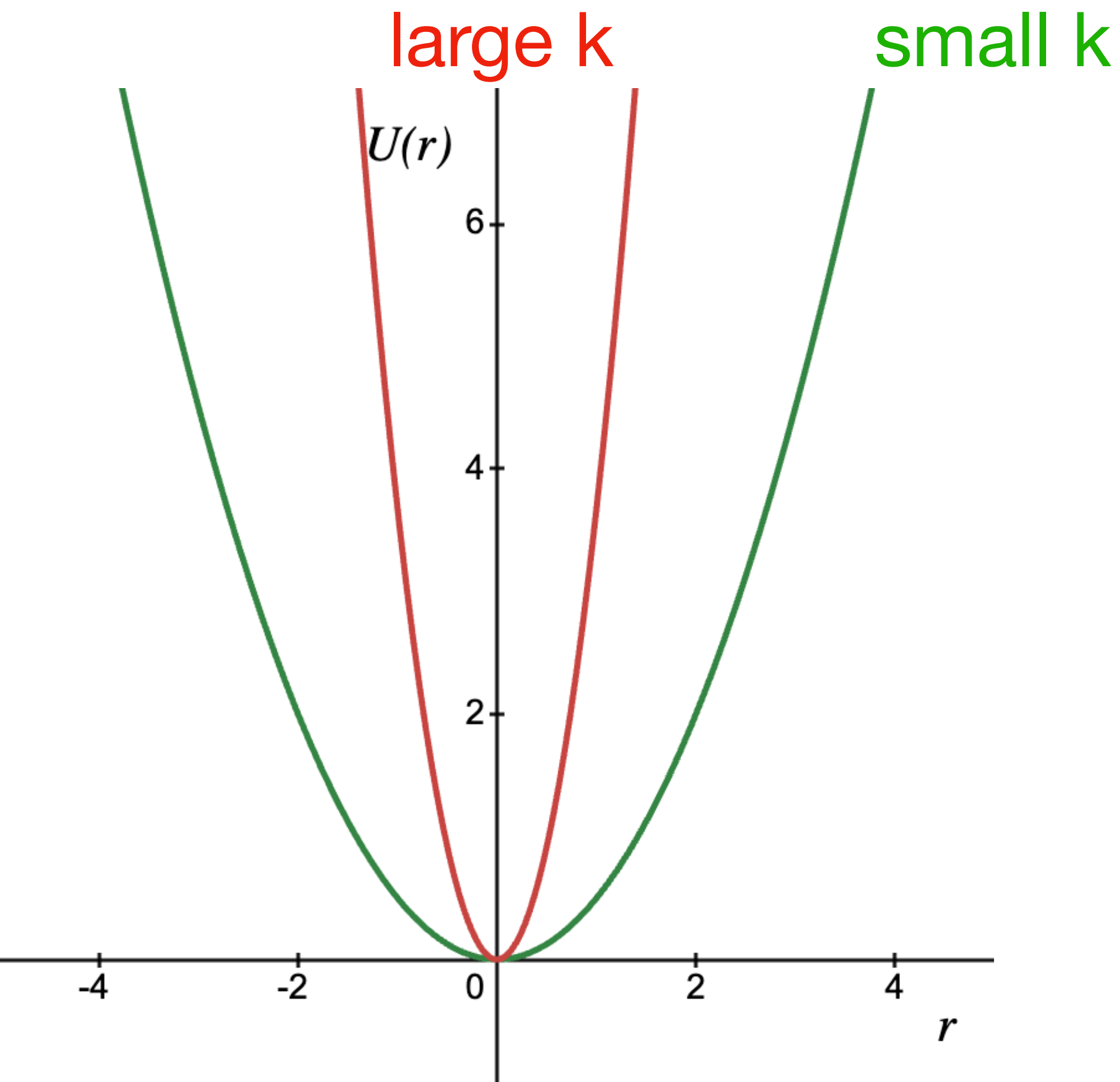
- Increased reactivity at the gold surface
- Highly polarizable due to sea of electrons
- Epitaxial growth and strong interactions with other gold atoms

How can we understand and visualize the molecular interactions of gold with biological materials?



What is a molecular dynamics simulation?

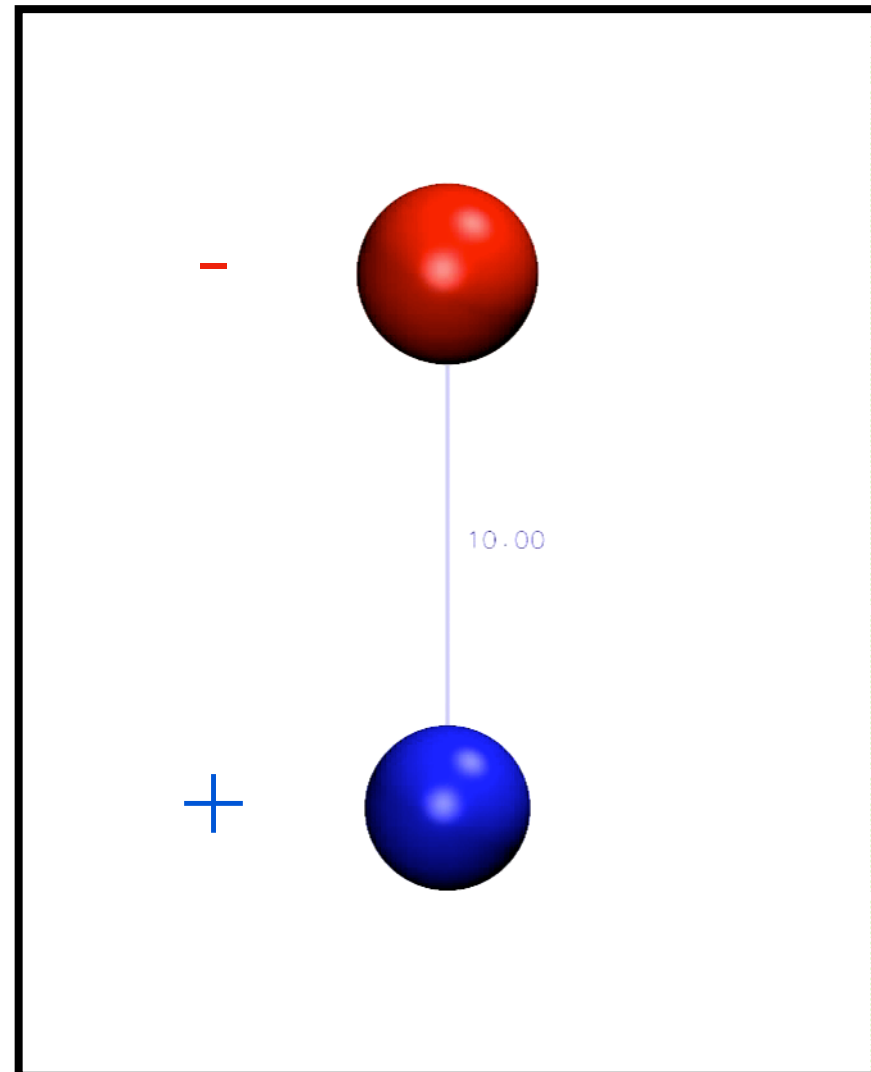
$$U(r) = \frac{1}{2}kr^2 \quad \text{Potential Energy}$$



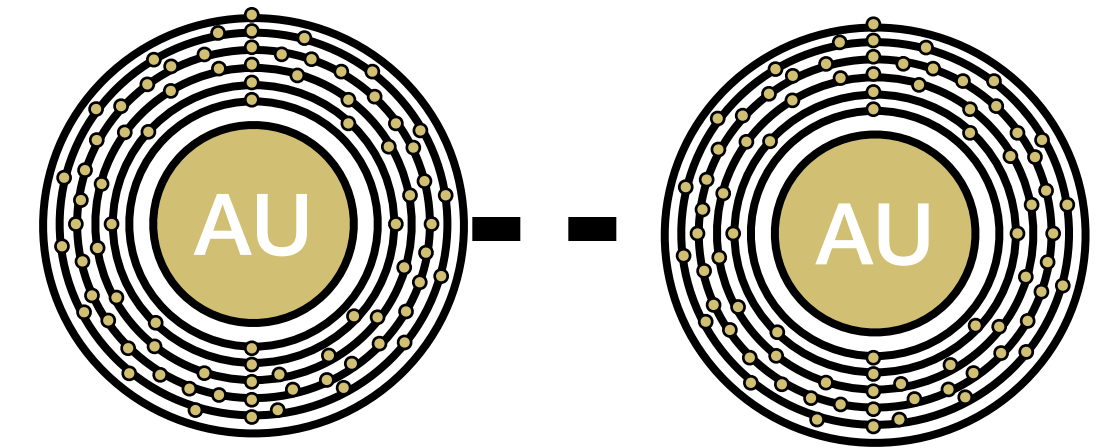
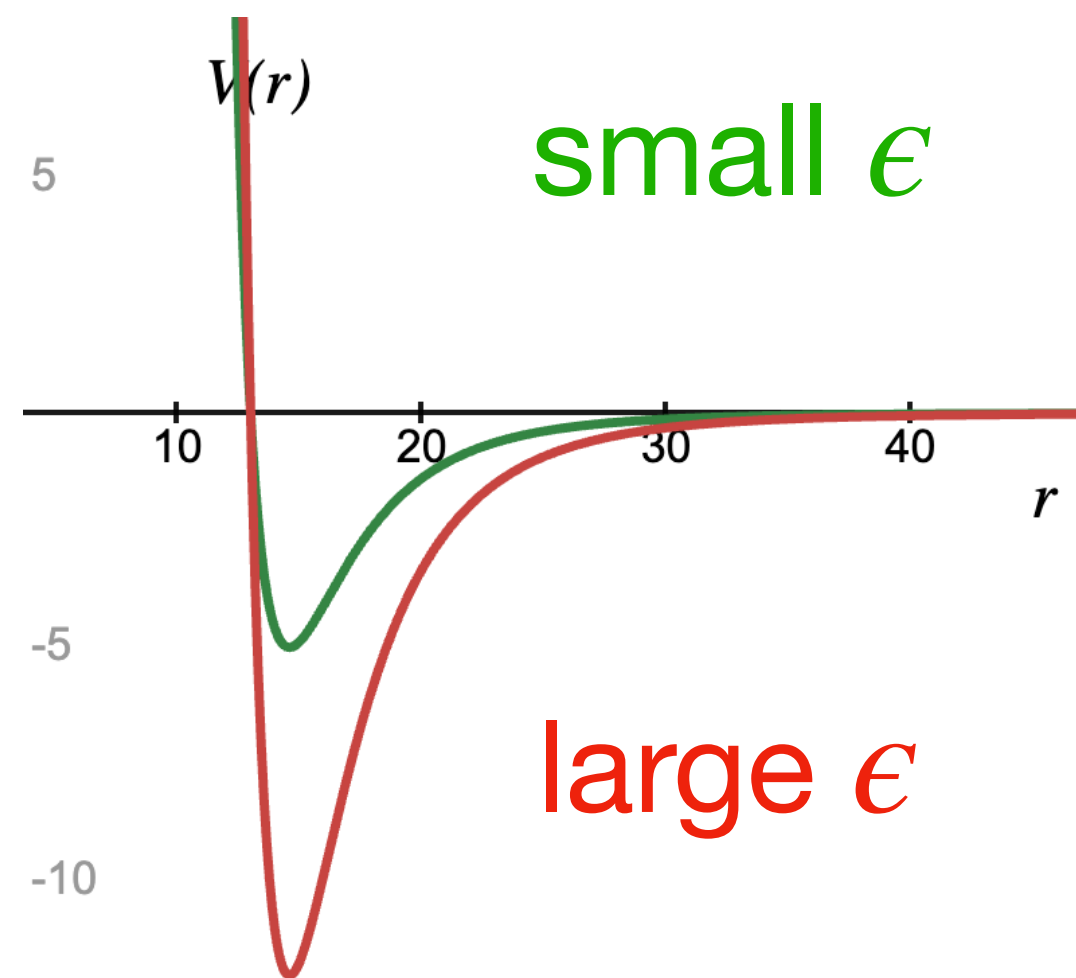
- We can use **potential energy** functions to determine a **molecules motion**
- Predict the movement of molecules

We describe systems through potential energy functions (ForceField)

Lennard Jones
& Electrostatic



$$V_{LJ}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$



- Lennard Jones controls gold-gold interaction strength
- INTERFACE force field

Resolution of molecular models

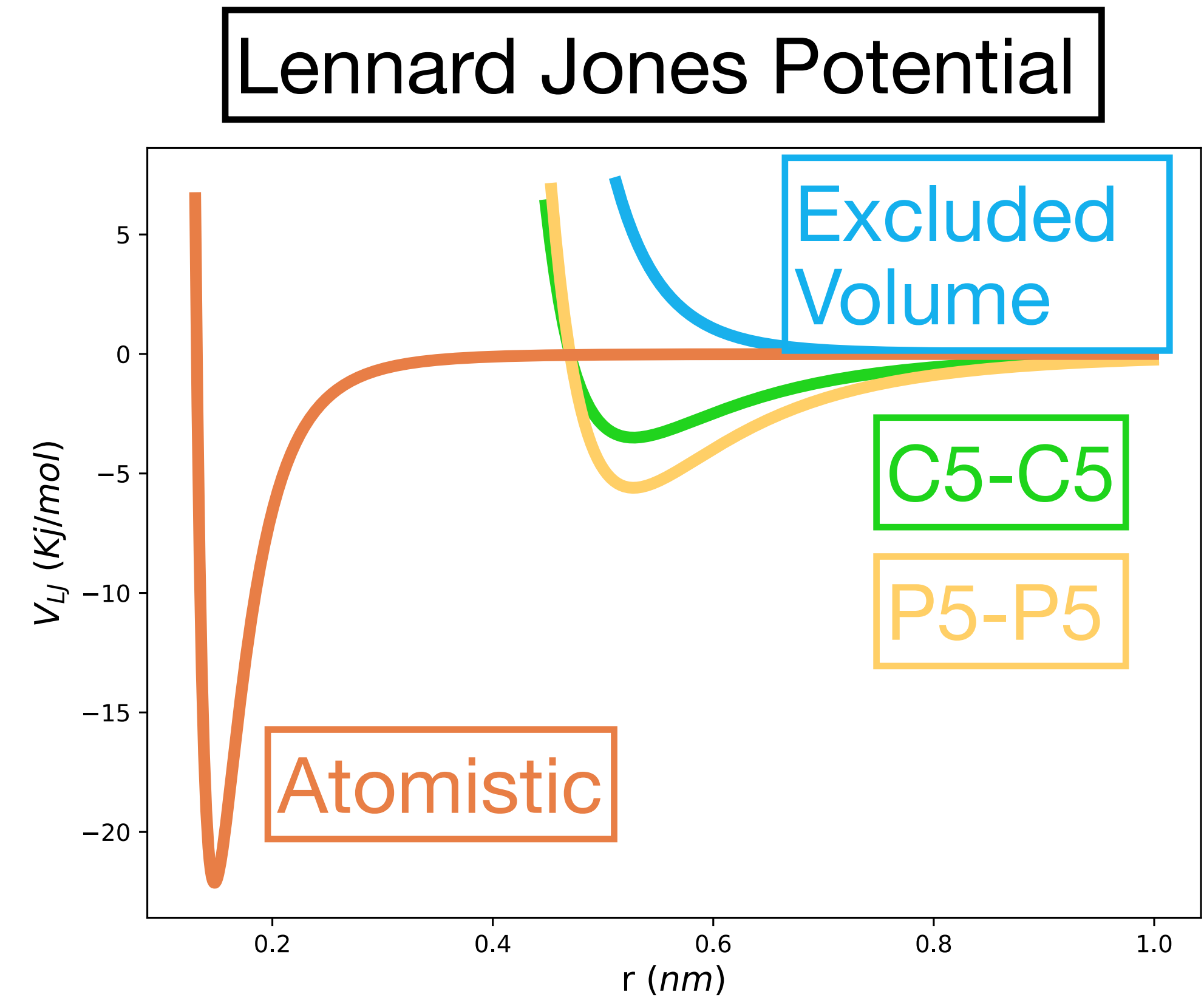
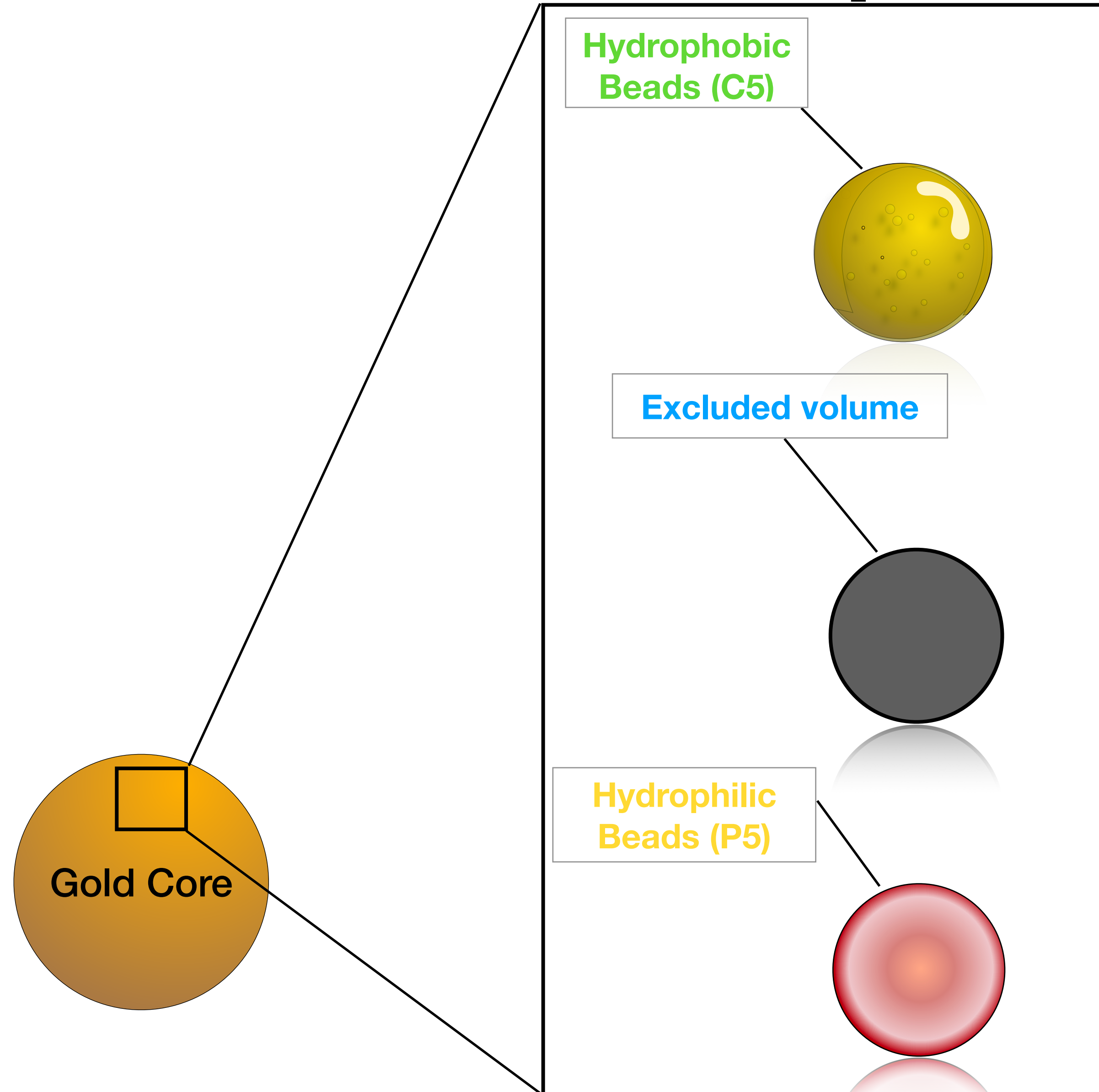
Atomistic
Models

Coarse-grained
Models

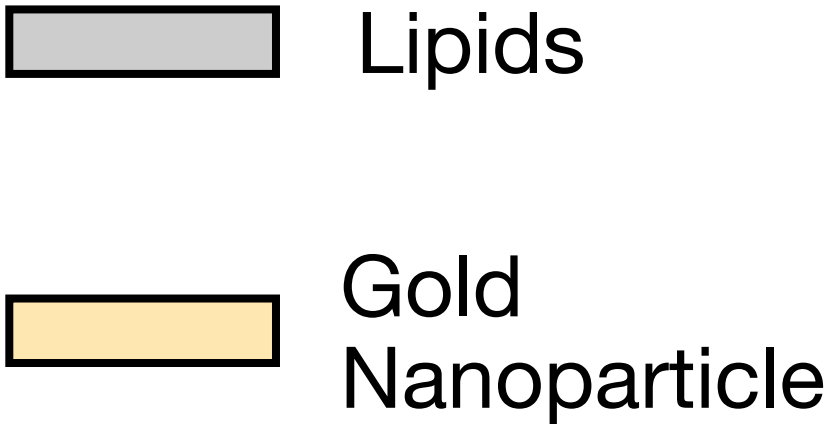
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How well can we reproduce gold in martini?

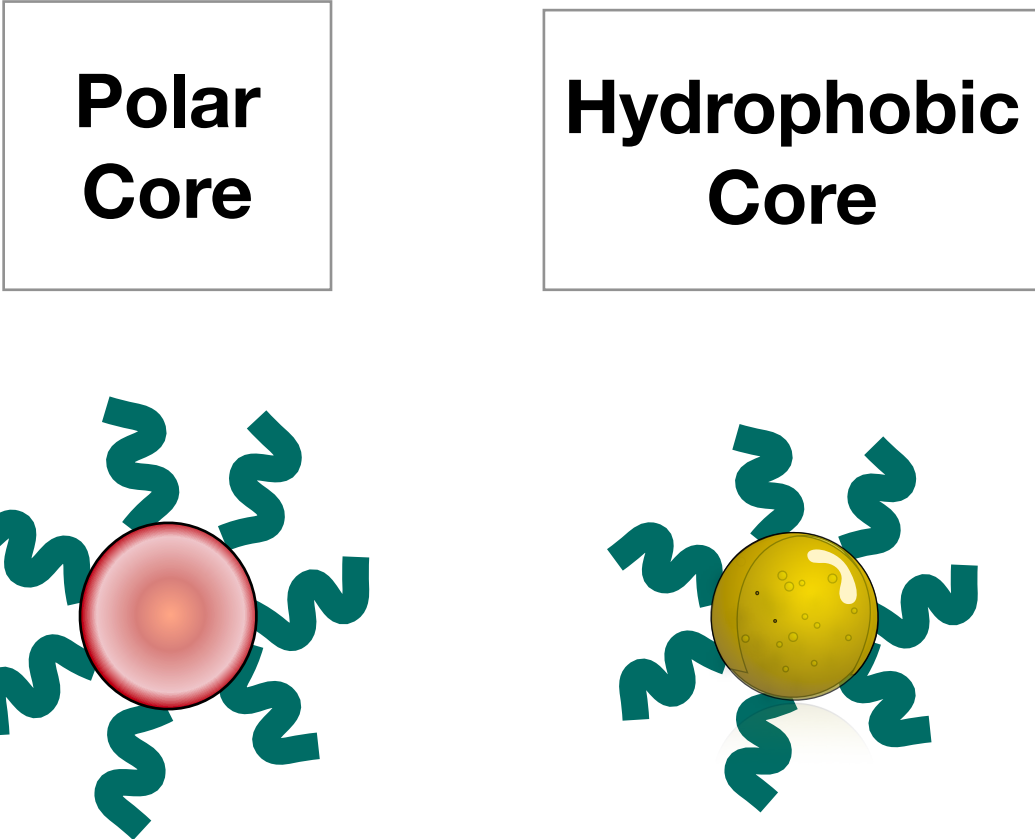


Gold nanoparticles interact differently depending on the coarse-grained gold core model



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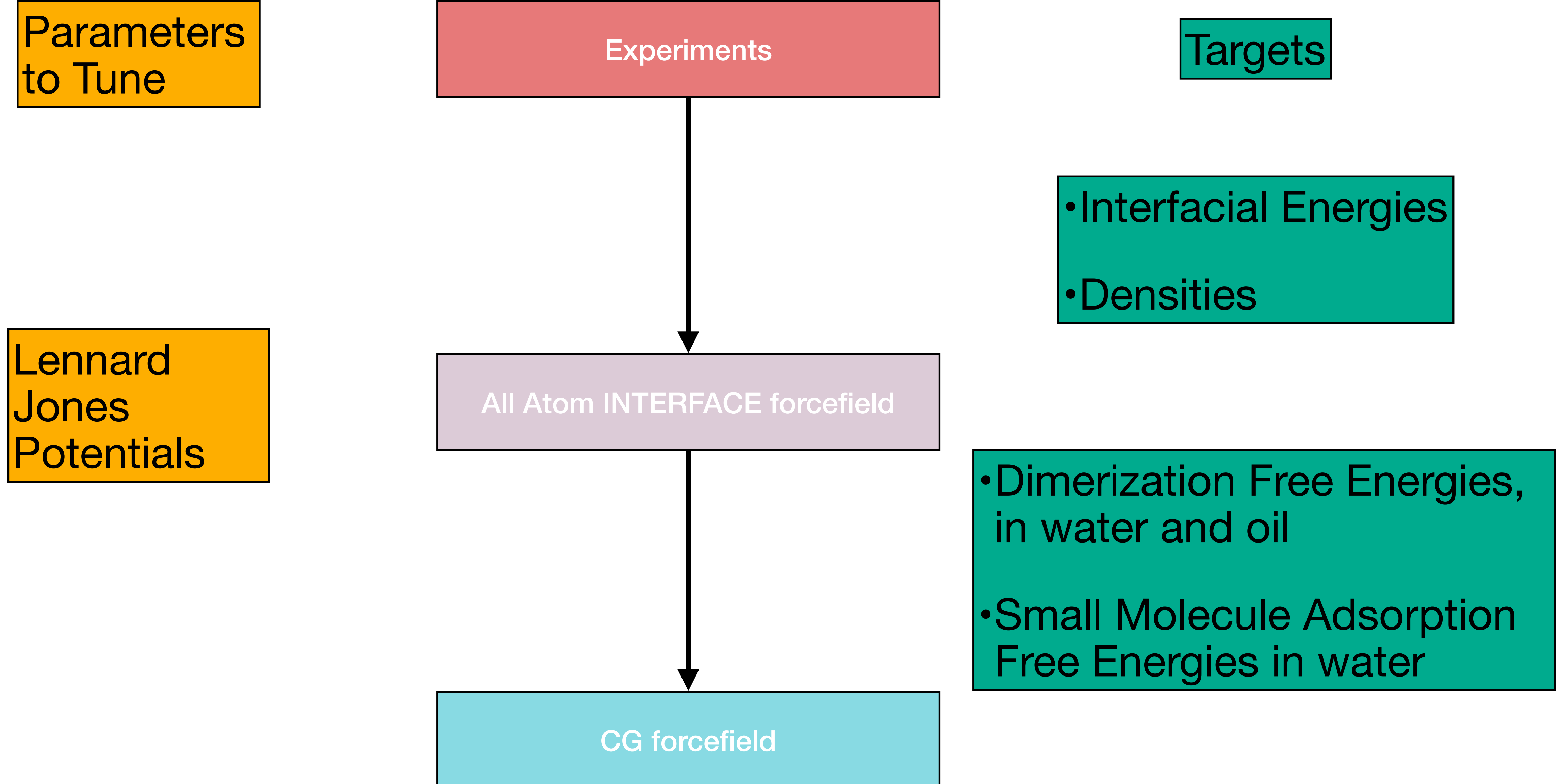
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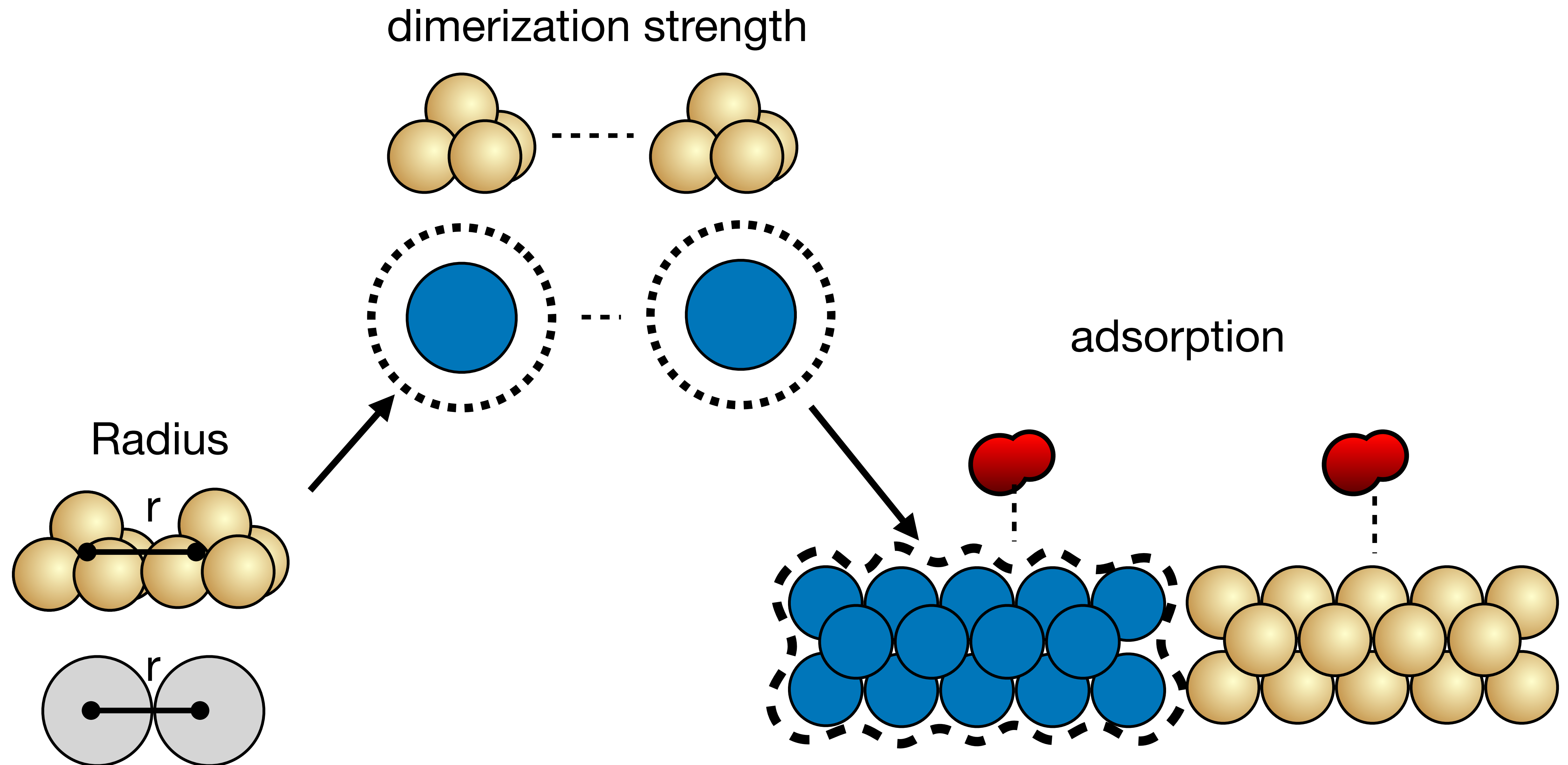
Our goal is to develop a coarse-grained gold bead that meets several criteria:

- Preserves strong gold-gold interactions
- Accurately reproduce atomistic and experimental data for small molecule adsorption
- Compatibility with a transferable biological force field (Martini)

Using All-Atom models as a base for coarse grained



The roadmap to developing gold

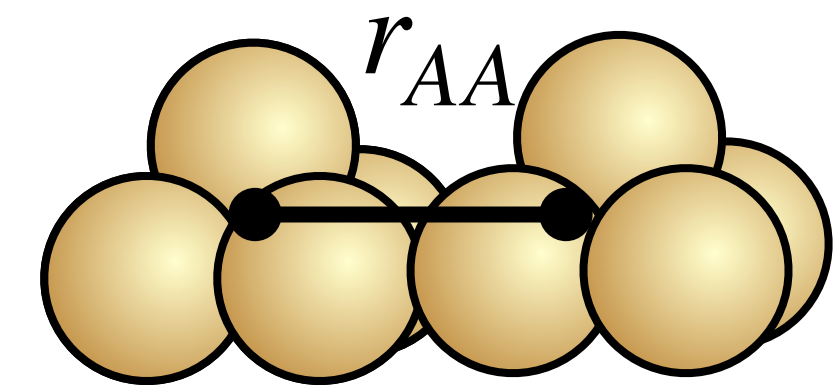


Validation and Model Refinement

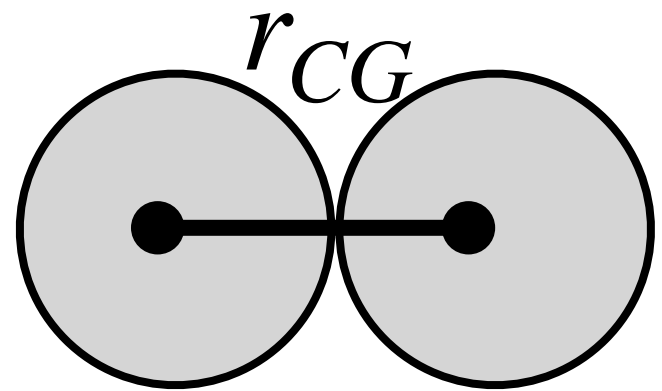
The roadmap to developing gold

r = global minimum

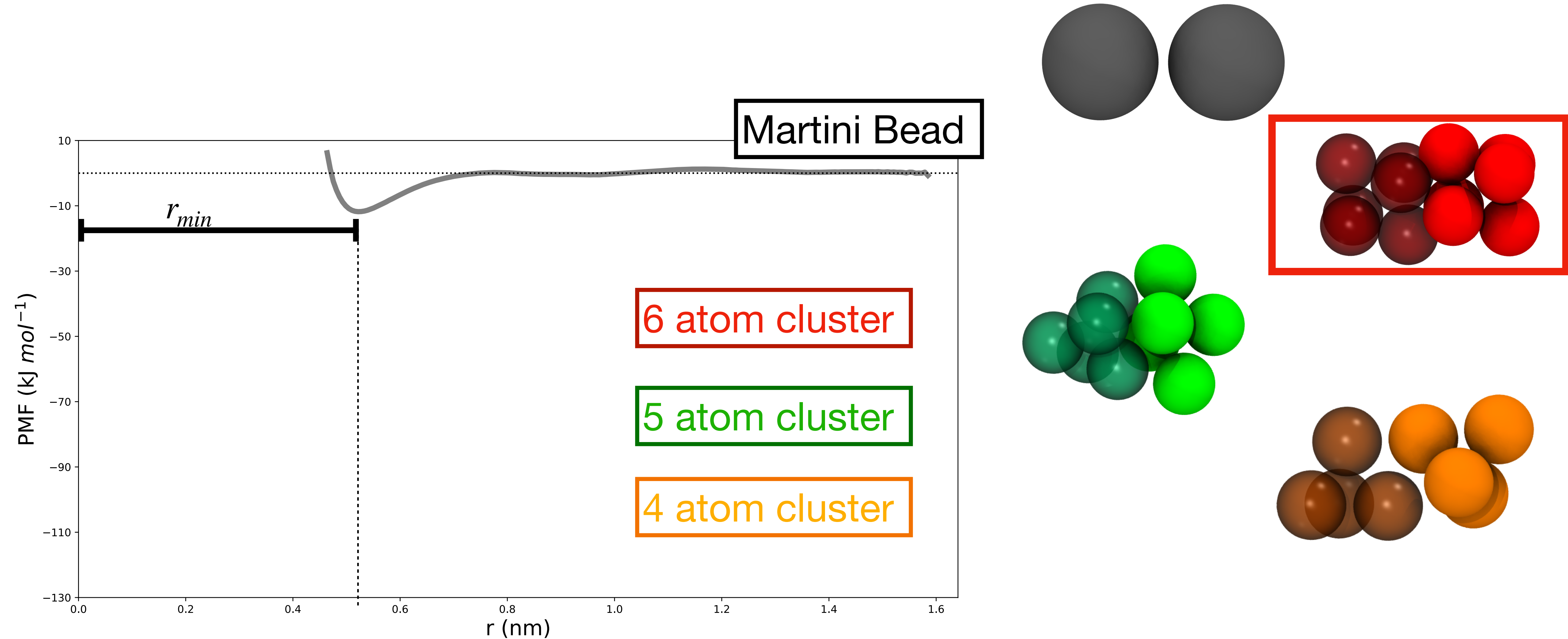
$$r_{cg} = r_{AA}$$



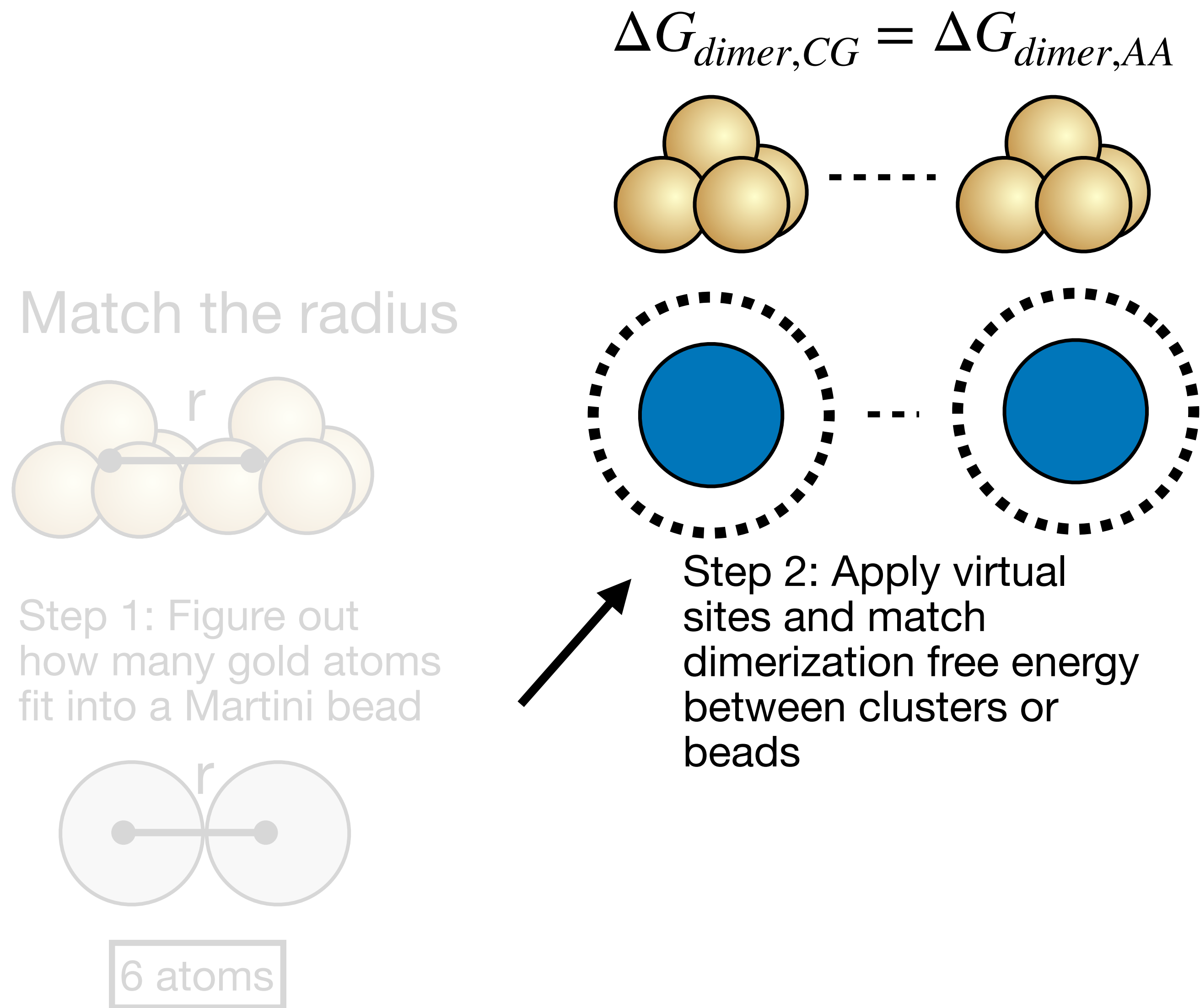
Step 1: Figure out
how many gold atoms
fit into a Martini bead



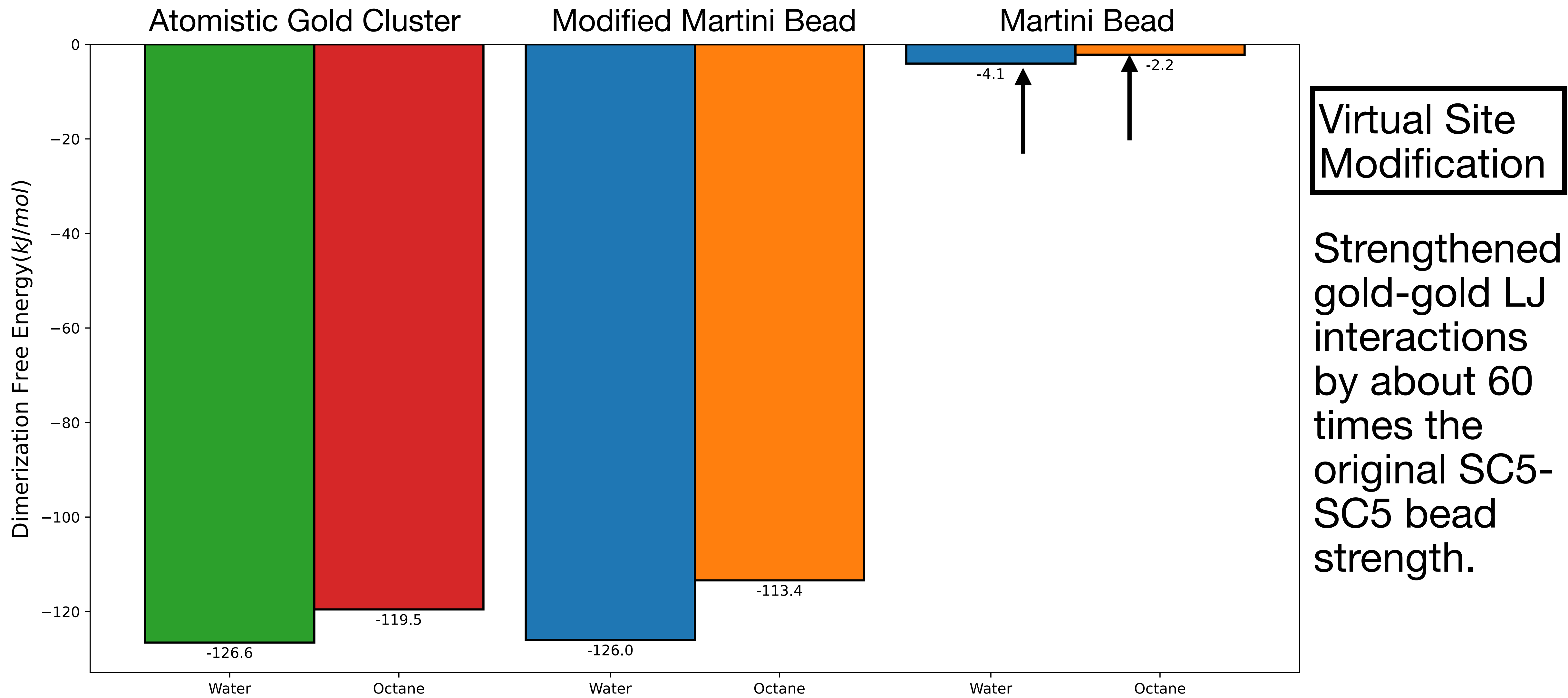
Determining the size of a gold bead



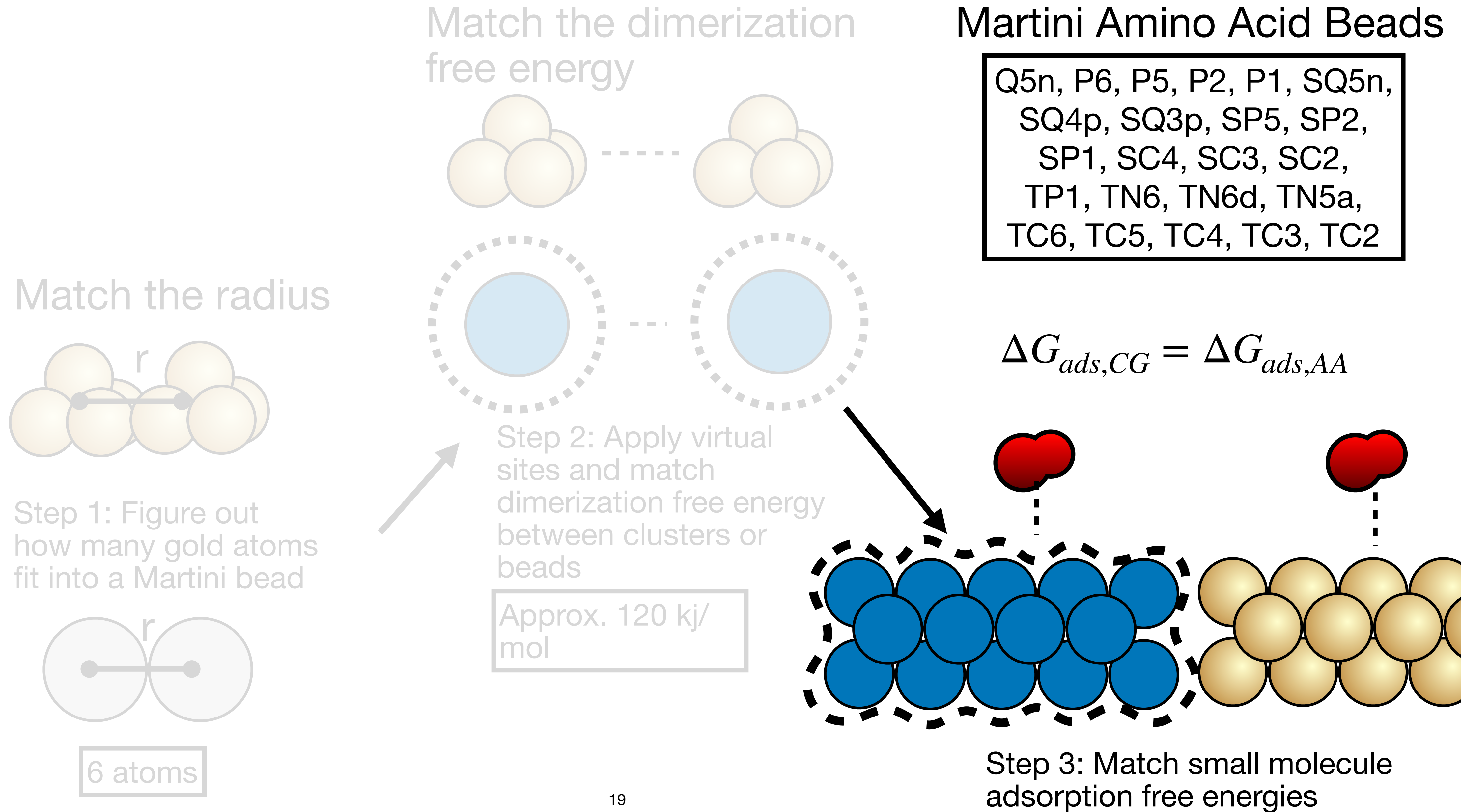
The roadmap to developing gold



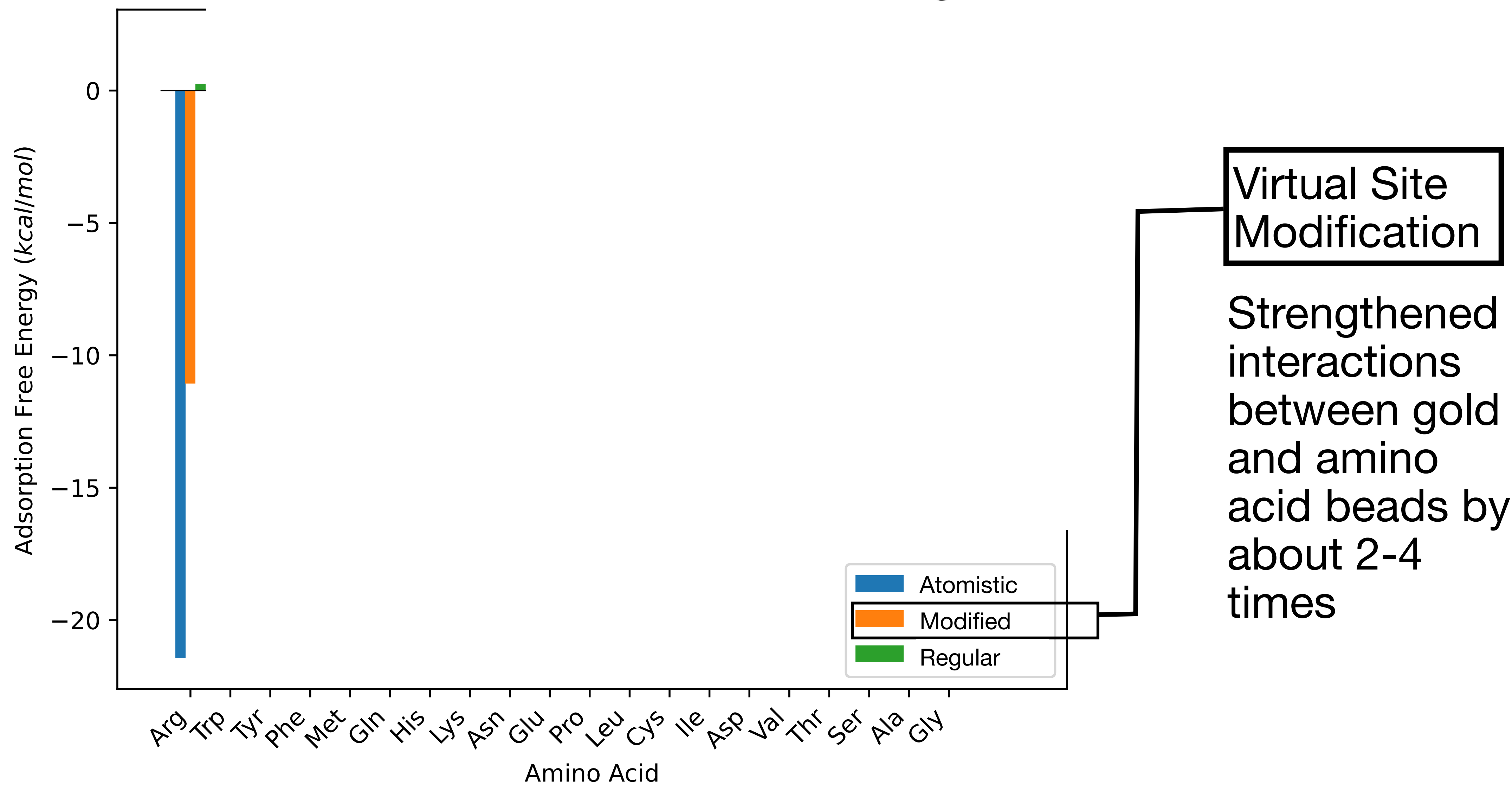
Matching the dimerization free energy between AA gold cluster and virtual site bead



The roadmap to developing gold



Matching adsorption free energies of amino acids between AA and virtual site gold slabs



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Summary & model refinement

- Present a method for developing Martini metal parameters that relies on free energy comparisons to atomistic simulation
- Present a Martini model for gold that:
 - Matches 6 atom cluster size
 - Uses a virtual site as a base to modify LJ potentials
 - Matches dimerization free energies of atomistic gold clusters
 - Incorporates polarization effects into interactions with small molecules
- Further refinement of the model will include:
 - * Adsorption free energies for Halogens, other charged molecules, etc...
 - * Extrapolate interactions to the entire Martini Matrix
 - * Comparison to experimental data

Acknowledgements

Image was here

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- Dr. Grace Brannigan



National
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Foundation

Questions

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